

HW6: Explainable AI

The following homework problems require you to make several implementation choices (neighborhood size and sampling approach in LIME or number of samples and approach to predicting with missing feature values in Shapley values). Any common or otherwise reasonable choice is OK, but you should be able to justify it and, even better, understand possible alternatives.

Part 1 (Grades 6–7): LIME

Implement the Local Interpretable Model-Agnostic Explanations (LIME) approach for tabular data. The LIME algorithm should be your own implementation and you should not rely on an existing LIME libraries. However, you may use a library implementation of Lasso regression as the local surrogate model.

You are provided with a dataset (`toydataset.csv`) with two continuous features x_1 , x_2 , one categorical feature x_3 , and a continuous target variable y . Train a model of your choice on the dataset (you can use a library for that). Then use your implementation of LIME to explain the model's prediction for the following samples:

x_1	x_2	x_3
0.4	-0.4	1
0.2	-0.4	1
0.4	-0.4	2
0.4	0.2	2

Part 2 (Grades 8–9): Shapley values

Implement Shapley values for explaining individual predictions of the model trained in Part 1. The implementation should be your own and you should not rely on an existing SHAP/Shapley libraries. Assume that the features are independent.

Use your implementation to explain the same samples as in Part 1.

Part 3 (+0, +0.5, +1.0 Grade):

Polish your report (language, plots, tables, structure, everything counts).

General

- In addition to these instructions, the homework includes a `toydataset.csv` file.

- Submit a PDF report (all parts combined into a single PDF; no more than 1 page per part!) and the code.
- Feel free to use LLMs and collaborate with others, but keep in mind that our goal is to understand what we are doing and not merely to do. Your work will be graded based on your understanding of your code, report, and the subject matter in general.